

Robotics Communication Protocols

Quick reference note — focus on "when to use what"

Definition: A set of rules that defines how data is transmitted and received between devices in a robotic system — like a common language all components agree to speak.

LAYER 1 — WIRED INTERNAL (INSIDE THE ROBOT BODY)

Wired

MCU ↔ Sensors / Actuators / Modules

Used between Arduino/STM32/ESP32 and the components directly connected to it.

UART

Simple serial

Use when: **only 2 devices talking**

Ex: Arduino ↔ GPS, Bluetooth, PC

I2C

2 wires, many devices

Use when: **multiple sensors, fewer pins**

Ex: IMU + LCD + Compass together

SPI

Fastest wired

Use when: **high-speed data needed**

Ex: Drone IMU, SD card, fast encoder

CAN Bus

Industrial grade

Use when: **big robot / noisy environment**

Ex: Car, robot arm, factory system

LAYER 2 — WIRELESS EXTERNAL (ROBOT ↔ OUTSIDE WORLD)

Wireless

Robot ↔ Phone / Laptop / Other Robots

Used to control or communicate with the robot remotely.

Wi-Fi

Long-range wireless

Use when: **wireless + long range control**

Ex: Mobile app ↔ Robot, ESP32 car

Bluetooth

Short-range wireless

Use when: **phone control, short range**

Ex: Android app ↔ Robot car

Ethernet

High-speed wired network

Use when: **huge data (LiDAR, camera)**

Ex: ROS PC ↔ Robot, Robot ↔ Robot

LAYER 3 — SOFTWARE MIDDLEWARE

Software

Intra-robot program communication

When multiple programs (camera node, navigation node, motor node) inside the robot need to share data — ROS acts as the translator.

ROS / ROS 2

Robot Operating System

Use when: **complex multi-node robot software**

Hides hardware, manages message passing between nodes

QUICK DECISION TABLE

Situation / Question	Use	Why
2 devices only (GPS, Bluetooth module)	UART	Simple, easy
...	...	...

Many sensors, few pins (IMU + display + compass)	I2C	2 wires, multi-device
High-speed sensor (drone flight controller)	SPI	Fastest wired
Industrial robot / car / factory	CAN Bus	Noise resistant, reliable
ROS-based system / LiDAR / camera stream	Ethernet	High speed, network
Wireless control, long range (laptop/phone)	Wi-Fi	Wireless, internet-type
Wireless control, short range (phone app)	Bluetooth	Cheap, easy, nearby
Multiple robot programs sharing data	ROS	Software middleware

EXAM SHORTCUT — 1-LINE MEMORY

UART → Simple (2 devices)	I2C → Many sensors, less wires	SPI → Fastest (embedded)
CAN → Industrial / vehicle	Ethernet → Huge data + network	Wi-Fi → Wireless long range
Bluetooth → Wireless short range	ROS → Software message bus	

এই note-এ সব কিছু একসাথে রাখা হয়েছে — definition থেকে শুরু করে layer-wise breakdown এবং সবচেয়ে গুরুত্বপূর্ণ "কখন কোনটা ব্যবহার করব" সেটা quick decision table-এ। Exam-এর আগে শুধু শেষের দুটো section দেখলেই হবে।

er baire kicu pora laghbe?

তোমার notes-এ যা আছে তা দিয়ে এই chapter-এর 70-80% কভার হয়ে যাবে।

তবে যদি একটু extra edge চাও, এই দুটো জিনিস দেখতে পারো:

1. Protocol-এর কিছু সংখ্যা মনে রাখা — যেমন I2C-তে max কতটা device connect করা যায়, UART-এর baud rate, SPI-এর speed। Viva-তে কেউ জিজ্ঞেস করলে কাজে আসে।

2. ROS সম্পর্কে একটু বেশি — ROS-এ "node", "topic", "publish/subscribe" মডেল কীভাবে কাজ করে, এটা অনেক university-তে আলাদা করে জিজ্ঞেস করে।

এই দুটো বাদ দিলে তোমার notes থেকে যা পড়েছ সেটাই যথেষ্ট। Exam কবে?

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